

Scalability & Future Plan

Date	30 September 2026
Team ID	PER206AIFITTRACK
Project Name	AI-FitTrack-API
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations vs Production Scalability Strategy:

Scalability Dimension	Current State (Local MVP)	Production Scalability Architecture
Concurrent User Traffic	Single Node.js process on Port 5000 handling up to 500 RPS	Deploy across multi-core Node.js cluster via PM2; Docker container instances behind AWS ALB / Nginx reverse proxy.
Data Persistence & Volume	Local MongoDB service (Port 27017) with compound indexing on userId/date	Migrate to MongoDB Atlas Multi-Region Replica Sets with automated horizontal sharding on userId partition key.
Generative AI Throughput	Synchronous REST requests to Google Gemini 1.5 Flash endpoint	Implement Redis caching for common workout routine prompts and asynchronous BullMQ job queues for heavy batch analytics.
Security & Rate Limiting	In-memory CORS and custom authMiddleware JWT verification	Implement express-rate-limit (100 req/min per IP), Helmet.js security headers, and AWS WAF edge firewall protection.

Future Product Roadmap Milestones:

Roadmap Milestone	Target Horizon	Planned Capabilities & Technical Scope
Phase 1: React Native Mobile App	Q1 2027	Cross-platform iOS/Android frontend connecting to existing RESTful endpoints with offline workout caching.
Phase 2: IoT Wearable Integration	Q2 2027	Bi-directional telemetry sync with Apple HealthKit, Google Health Connect, and Garmin smartwatches.
Phase 3: Multimodal Vision AI	Q3 2027	Gemini Vision AI meal recognition computing macronutrients from plate photos and posture analysis during lifts.